MCQ Answer Key

1) c) 2:1

Explanation:

$$\sigma_\theta = \frac{pd}{2t} \qquad \sigma_l = \frac{pd}{4t}$$

$$\frac{\sigma_\theta}{\sigma_l} = \frac{\frac{pd}{2t}}{\frac{pd}{4t}} \Rightarrow \frac{\sigma_\theta}{\sigma_l} = \frac{2}{1}$$

2) c) $pd/8t$

Exp:

$$\tau_{max} = \frac{\sigma_h - \sigma_l}{2} = \frac{\frac{pd}{2t} - \frac{pd}{4t}}{2} = \frac{pd}{8t}$$

3) b) $\leq 1/15$

Exp:

$\frac{t}{d} \leq \frac{1}{15}$ for thin cylinders $\qquad \frac{t}{d} > \frac{1}{15}$ for thick cylinders

4) c) 2%.

Exp: $\sigma_L = \frac{pr}{4t} \qquad \sigma_\theta = \frac{pr}{2t}$

Taking logarithms and differentiating

$$\frac{d\sigma}{\sigma} = \frac{dr}{r} - \frac{dt}{t} = 1 - (-1) = 2\%.$$

5) c) Inner surface

Reason: In thick cylinders, the hoop stress reaches its peak at the inner radius and decreases towards the outer radius.

6) b) Boundary conditions of radial stress at inner and outer radii

Reason: Constants are determined using radial stress boundary conditions at the inner and outer radii

7) a) $5p/3$

Explanation:

$r_i = d/2 \qquad r_o = 2d/2$

$$\sigma_{h,max} = p \cdot \frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} = p \cdot \frac{d^2 + (d/2)^2}{d^2 - (d/2)^2} = p \frac{\frac{5d^2}{4}}{\frac{3d^2}{4}}$$

$$= \frac{5p}{3}$$

8) c) Both axial and hoop stress, no shear

Reason: The liquid pressure induces circumferential stress, while the weight of the container hanging from above induces axial tensile stress

9) c) $-p_i$

Reason:

Radial stress is compressive in nature

10) b) -0.1%

Explanation:

$\varepsilon_h = 0.1\%$

$$\varepsilon_t = -\frac{\nu}{1-\nu}(\varepsilon_h + \varepsilon_l)$$

for $\nu = \frac{1}{3}$

$$\varepsilon_t = -\varepsilon_h = -0.1\%$$

11) b) Thin cylinders

12) b) Uniform

Reason: Due to small wall thickness, the variation of stress across the thickness is negligible and assumed uniform.

13) b) Greater than 1/15

Exp: $\frac{t}{d} \leq \frac{1}{15}$ for thin cylinders $\frac{t}{d} > \frac{1}{15}$ for thick cylinders

14) b) Maximum at inner radius, zero at outer radius.

Reason: At inner radius, radial stress is equal to internal pressure. At the unpressurized outer radius, radial stress drops to zero.

15) b) Reduce the resultant hoop stress in the combined cylinder.

Explanation: Shrinking creates compressive initial stresses at the inner cylinder, which counteracts the high hoop stresses caused by internal pressure.

16) b) $pd/4t$

Explanation: In sphere,

$$\sigma_\theta = \frac{pd}{4t}$$

17) c) Stress varies uniformly across the thickness like a thin cylinder.

18) b) $2\varepsilon_h + \varepsilon_l$

Explanation:

$$\varepsilon_V = \frac{\Delta V}{V} = 2\varepsilon_h + \varepsilon_l$$

↑ volumetric strain

19) b) Three times the hoop strain

Reason: $\varepsilon_x = \varepsilon_y = \varepsilon_z = \varepsilon_h$

$$3\varepsilon_h = \varepsilon_V$$

20) c) Both boundary conditions.

Explanation:

Two boundary conditions are required to solve the two unknown constants A and B

1) Given:

$r = 100\ mm$

$d = 200\ mm$

$p = 10\ MPa = 10\ N/mm^2$

$\sigma_\theta = 100\ MPa$

To find:

Thickness, $T = ?$

Solution:

for hoop stress

$$\sigma_\theta = \frac{pd}{2t} \longrightarrow 100 = \frac{10(200)}{2(t)}$$

$$t = \frac{10(200)}{2(100)}$$

$t = 10mm$, option a

2) Given:

$d = 100mm$

$t = 5mm$

$p = 10\ MPa = 10\ N/mm^2$

$T = 2000\ Nm = 2000 \times 10^3\ Nmm$

To find:

Magnitudes of principal stresses,

[It can be found by calculating the shear and normal components.]

Finding Hoop & Axial stress :

$$\sigma_\theta = \frac{Pd}{2t} \qquad \sigma_L = \frac{Pd}{4t}$$

$$= \frac{10(100)}{2(5)} \qquad = \frac{10(100)}{4(5)}$$

$$\sigma_\theta = 100 \qquad \sigma_L = 50$$

To find shear stress:

$$\frac{T}{J} = \frac{\tau}{R} \implies \tau = \frac{RT}{J} \longrightarrow (1)$$

$$J = \frac{\pi}{32}(D^4 - d^4)$$

$$J = \frac{\pi}{32}(0.11^4 - 0.1^4)$$

$D = d + 2t$

$= 100 + 2(5)$

$= 110\,mm = 0.11m$

$R = 55mm = 0.055m$

$J = 4.556 \times 10^{-6}\ m^4$, sub in (1)

$$\tau = \frac{0.055(2000)}{4.556 \times 10^{-6}}$$

$$\tau = 24.14\ MN/m^2$$

By calculating the shear and normal components

$$\sigma_{1,2} = \frac{\sigma_\theta + \sigma_L}{2} \pm \sqrt{\left(\frac{\sigma_\theta - \sigma_L}{2}\right)^2 + \tau^2}$$

$$= \frac{100+50}{2} \pm \sqrt{\left(\frac{100-50}{2}\right)^2 + (24.14)^2}$$

$$= \frac{150}{2} \pm \sqrt{\left(\frac{50}{2}\right)^2 + 24.14^2}$$

$$= 75 \pm 34.75$$

$$\sigma_{1,2} = 40.25, \ 109.75 \quad \text{option a}$$

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3) Given:

$$\nu = \frac{1}{3} = 0.33$$

$$\varepsilon_\theta = 0.1\%$$

To find:

Percentage change of thickness strain

Solution:

w.k.t for hoop strain $\left[\sigma_L = \frac{\sigma_\theta}{2}\right]$

$$\varepsilon_\theta = \frac{\sigma_\theta}{E} - \nu\frac{\sigma_L}{E} \longrightarrow \frac{\sigma_\theta}{E} - \frac{1}{3}\left(\frac{1}{2}\right)\frac{\sigma_\theta}{E}$$

$$\varepsilon_\theta = \frac{\sigma_\theta}{E}\left(\frac{5}{6}\right) \longrightarrow (1)$$

w.k.t thickness strain

$$\varepsilon_t = \frac{-\nu}{E}(\sigma_\theta + \sigma_L)$$

$$= \frac{-1}{3E}\left(\sigma_\theta + \frac{\sigma_\theta}{2}\right) = \frac{-1}{E}\left(\frac{\sigma_\theta}{2}\right) \longrightarrow (2)$$

(1) ÷ (2)

$$\frac{\varepsilon_\theta}{\varepsilon_t} = \frac{\frac{\sigma_\theta}{E}\left(\frac{5}{6}\right)}{\frac{-1}{E}\left(\frac{\sigma_\theta}{2}\right)}$$

$$\frac{\varepsilon_o}{\varepsilon_t} = \frac{\sigma_o}{E}\left(\frac{5}{6}\right)\left(\frac{-E}{1}\right)\left(\frac{2}{\sigma_o}\right)$$

3

$$\frac{\varepsilon_o}{\varepsilon_t} = \frac{-5}{3}$$

$$\varepsilon_t = -0.1 \times \frac{3}{5} \qquad [\varepsilon_o = 0.1\%]$$

$$\varepsilon_t = -0.06\%$$

It falls between -0.062 & -0.058%,

so option a

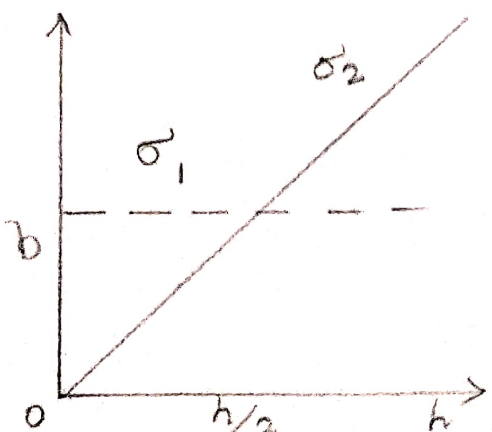
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4) It is given that the vessel is suspended from the top.

* The total tensile axial force at any cross section y remains constant (σ_1) since the vessel is suspended from the top and entire downward weight is transmitted uniformly up through the cylinder walls.

* Since the vessel is suspended from the top, At the base pressure will be maximum. At the top pressure will be zero.

* So pressure increases from top to the base region (h) for σ_2 so option c is correct



5) Given:

$a = 60°$

$t = 10\,mm$

$d = 2m = 2000\,mm$

$P = 200\,KPa = 0.2\,MPa = 0.2\,N/mm^2$

To find:

Normal stress

Solution:

$$\sigma_\theta = \frac{Pd}{2t} = \frac{0.2(2000)}{2(10)} = 20\,N/mm^2$$

$$\sigma_L = \frac{Pd}{4t} = \frac{0.2(2000)}{4(10)} = 10\,N/mm^2$$

Normal stress

$$\sigma_n = \sigma_\theta \cos^2 a + \sigma_L \sin^2 a$$

$$\sigma_n = 20 \cos^2 60 + 10 \sin^2 60$$

$$= 20(0.25) + 10(0.75)$$

$$= 5 + 7.5$$

$$\sigma_n = 12.5$$

Falls under 12.3 – 12.7, so option a

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6) Given:

$d = 3\,m = 3000\,mm$

$t = 15\,mm$ $\qquad \varepsilon_\theta = 0.00034$

$\nu = 0.3$

$E = 210\,GPa = 210 \times 10^3\,N/mm^2$

To find:

Pressure $P = ?$

Solution:

w.k.t for hoop strain

$$\varepsilon_\theta = \frac{\sigma_\theta}{E} - \nu\frac{\sigma_L}{E}$$

$$= \frac{1}{E}(\sigma_\theta - \nu\sigma_\ell)$$

$$= \frac{1}{E}\left(\sigma_\theta - \frac{\nu\sigma_\theta}{2}\right) \rightarrow \frac{\sigma_\theta}{E}\left(1 - \frac{\nu}{2}\right)$$

$$= \frac{pd}{2tE}\left(1 - \frac{\overset{0.3}{\nu}}{2}\right)$$

$$0.00034 = \frac{P(3000)}{2(15)(210\times10^3)}(0.85)$$

$$P = 0.84\ MPa$$

$$P = 840\ KPa \qquad \text{option a}$$

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7) Given:

$R = 1m = 1000mm$

$d = 2m = 2000mm$

$\rho = 1000\ kg/m^3$

$g = 10\ m/s^2$

given mid depth, $y = 1000mm$ (or) $1m$

To find: Axial & circumferential stress

σ_L & $\sigma_\theta = ?$

w.k.t pressure at depth

$$P = \rho g y = 1000 \times 10 \times 1$$

$$= 10000 \text{ Pa}$$

$$= 0.01 \text{ MPa}$$

$$= 0.01 \text{ N/mm}^2$$

$$\text{Hoop stress } (\sigma_0) = \frac{pd}{2t} = \frac{0.01(2000)}{2(1)}$$

$$= 10 \text{ N/mm}^2$$

$$\text{Axial stress } (\sigma_L) = 0.5\sigma_0 = 5 \text{ N/mm}^2$$

option a

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8) Given:

$R = 1 \text{ m} = 1000 \text{ mm}$

$d = 2 \text{ m} = 2000 \text{ mm}$

$\rho = 1000 \text{ kg/m}^3$

$g = 10 \text{ m/s}^2$

$E = 100 \text{ GPa} = 100 \times 10^3 \text{ N/mm}^2$

$\nu = 0.3$

To find:

Axial strain in the cylinder wall

Solution:

From question 7, $\sigma_0 = 10 \text{ N/mm}^2$

$\sigma_L = 5 \text{ N/mm}^2$

w.k.t axial strain

$$\varepsilon_L = \frac{\sigma_L}{E} - 2\frac{\sigma_o}{E}$$

$$= \frac{5 - 0.3(10)}{100000}$$

$$\varepsilon_L = 2 \times 10^{-5}$$

option a

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9) To find:

$\sigma_o : \sigma_L$

Solution:

w.k.t hoop stress

$$\sigma_\theta = \frac{Pd}{2t} \rightarrow (1)$$

axial stress

$$\sigma_L = \frac{Pd}{4t} \rightarrow (2)$$

$$(1) \div (2) \Longrightarrow \frac{\sigma_o}{\sigma_L} = \frac{\frac{Pd}{2t}}{\frac{Pd}{4t}}$$

$$\frac{\sigma_o}{\sigma_L} = \frac{Pd}{2t} \cdot \frac{4t}{Pd} \rightarrow \frac{\sigma_o}{\sigma_L} = \frac{2}{1} \rightarrow 2:1$$

option c

10) Given:

$P = 500\,KPa = 0.5\ N/mm^2$

$r = 2m = 2000\,mm$, $d = 4000\,mm$

$t = 10\,mm$

To find:

τ_{max}

Solution:

Maximum in plane shear stress is given by

$$\tau_{max} = \frac{\sigma_\theta - \sigma_L}{2}$$

$$= \frac{\frac{Pd}{2t} - \frac{Pd}{4t}}{2}$$

$$= \frac{\frac{0.5(4000)}{2(10)} - \frac{0.5(4000)}{4(10)}}{2}$$

$$= \frac{100 - 50}{2}$$

$$\tau_{max} = 25\ MPa$$

option a